

Web Programming Course

Student Group Project Assignments

Department of Computer Science and Engineering, UIU

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Technology Stack

Every project should be developed using (but not limited to):

- HTML
- CSS
- JavaScript
- PHP
- MySQL

General Requirements

All groups must implement the following.

- Secure Login and Registration
- Password hashing using `password_hash()`
- Minimum two user roles
- Responsive interface
- Client-side and Server-side validation
- Prepared Statements
- XSS prevention
- CRUD operations
- Search and filtering
- Dashboard with statistics
- Clean folder structure

Project 1: Campus Blood Donation Portal

Project Overview

A platform connecting blood donors and patients.

User Roles

- Administrator
- User

Functional Requirements

1. Register as donor
2. Request blood
3. Search donors
4. Donation history
5. Admin approval
6. Responsive user interface
7. Search and filter
8. Profile management
9. Dashboard

Suggested Database Tables

- users
- admin
- main module table(s)
- requests or transactions
- notifications

Unique Twist

Simple Challenge

Emergency requests stay pinned on top until fulfilled.

Optional Bonus

- Email-style notification inside the website
- Dark mode
- Export reports as PDF

Project 2: Parking Reservation System

Project Overview

Reserve parking slots inside campus.

User Roles

- Administrator
- User

Functional Requirements

1. Slot management
2. Book slot
3. Check-in
4. Check-out
5. Booking history
6. Responsive user interface
7. Search and filter
8. Profile management
9. Dashboard

Suggested Database Tables

- users
- admin
- main module table(s)
- requests or transactions
- notifications

Unique Twist

Simple Challenge

Three late departures temporarily disable booking for one day.

Optional Bonus

- Email-style notification inside the website
- Dark mode
- Export reports as PDF

Project 3: Pet Adoption Portal

Project Overview

Adoption management for animal shelters.

User Roles

- Administrator
- User

Functional Requirements

1. Pet listing
2. Search
3. Apply
4. Approve adoption
5. Success stories
6. Responsive user interface
7. Search and filter
8. Profile management
9. Dashboard

Suggested Database Tables

- users
- admin
- main module table(s)
- requests or transactions
- notifications

Unique Twist

Simple Challenge

Compatibility score based on user preferences.

Optional Bonus

- Email-style notification inside the website
- Dark mode
- Export reports as PDF

Project 4: Freelancer Marketplace

Project Overview

Simple freelance hiring portal.

User Roles

- Administrator
- User

Functional Requirements

1. Post jobs
2. Apply
3. Hire
4. Ratings
5. Portfolio
6. Responsive user interface
7. Search and filter
8. Profile management
9. Dashboard

Suggested Database Tables

- users
- admin
- main module table(s)
- requests or transactions
- notifications

Unique Twist

Simple Challenge

Automatic Trusted Freelancer badge after five successful jobs.

Optional Bonus

- Email-style notification inside the website
- Dark mode
- Export reports as PDF

Project 5: Campus Complaint System

Project Overview

Track complaints digitally.

User Roles

- Administrator
- User

Functional Requirements

1. Submit complaint
2. Track status
3. Assign staff
4. Announcements
5. Reports
6. Responsive user interface
7. Search and filter
8. Profile management
9. Dashboard

Suggested Database Tables

- users
- admin
- main module table(s)
- requests or transactions
- notifications

Unique Twist

Simple Challenge

Complaints unanswered for three days become High Priority automatically.

Optional Bonus

- Email-style notification inside the website
- Dark mode
- Export reports as PDF

Project 6: Digital Library

Project Overview

Reserve books online.

User Roles

- Administrator
- User

Functional Requirements

1. Book catalog
2. Search
3. Reserve
4. Borrow history
5. Fine calculation
6. Responsive user interface
7. Search and filter
8. Profile management
9. Dashboard

Suggested Database Tables

- users
- admin
- main module table(s)
- requests or transactions
- notifications

Unique Twist

Simple Challenge

Homepage displays Top 5 most borrowed books.

Optional Bonus

- Email-style notification inside the website
- Dark mode
- Export reports as PDF

Project 7: Neighborhood Help Portal

Project Overview

People request and provide help.

User Roles

- Administrator
- User

Functional Requirements

1. Create request
2. Volunteer
3. History
4. Notifications
5. Status updates
6. Responsive user interface
7. Search and filter
8. Profile management
9. Dashboard

Suggested Database Tables

- users
- admin
- main module table(s)
- requests or transactions
- notifications

Unique Twist

Simple Challenge

Each completed task increases a public Kindness Score.

Optional Bonus

- Email-style notification inside the website
- Dark mode
- Export reports as PDF

Project 8: Internship Portal

Project Overview

Students apply for internships.

User Roles

- Administrator
- User

Functional Requirements

1. Company dashboard
2. Internship posts
3. Applications
4. Application status
5. Profiles
6. Responsive user interface
7. Search and filter
8. Profile management
9. Dashboard

Suggested Database Tables

- users
- admin
- main module table(s)
- requests or transactions
- notifications

Unique Twist

Simple Challenge

Countdown timer for applications closing within three days.

Optional Bonus

- Email-style notification inside the website
- Dark mode
- Export reports as PDF

Project 9: Food Donation Platform

Project Overview

Restaurants donate excess food.

User Roles

- Administrator
- User

Functional Requirements

1. Food listings
2. Request food
3. Pickup confirmation
4. Donation history
5. Admin dashboard
6. Responsive user interface
7. Search and filter
8. Profile management
9. Dashboard

Suggested Database Tables

- users
- admin
- main module table(s)
- requests or transactions
- notifications

Unique Twist

Simple Challenge

Cards automatically change Green, Yellow or Red based on remaining freshness.

Optional Bonus

- Email-style notification inside the website
- Dark mode
- Export reports as PDF

Project 10: E-Commerce Marketplace

Project Overview

Develop a multi-vendor E-Commerce Marketplace where multiple sellers can create their own online stores and buyers can browse, purchase, and review products.

User Roles

- **Buyer**
- **Seller**
- **Administrator**

Functional Requirements

1. User Management System
2. Product Management
3. Order Processing System
 - Shopping cart with quantity modification.
 - Persistent shopping cart after user login.
 - Automatic total price calculation.
 - Support promotional discounts.
 - Place orders using selected shipping address.
 - Order tracking (Pending, Confirmed, Shipped, Delivered).
 - Order history.
 - Printable invoice page.
4. Communication System
 - Buyer-Seller private messaging.
 - Conversation history.
 - Notification for unread messages.
 - Product review system.
 - Five-star rating system.
 - Review moderation by administrators.
 - Average product rating display.
5. Administration Panel
 - User account management.
 - Seller verification.
 - Product approval and moderation.
 - Complaint management.
 - Transaction dispute management.
 - Marketplace analytics dashboard.

Suggested Database Tables

- users
- seller_profiles
- buyer_profiles
- categories
- products
- product_images
- shopping_cart
- cart_items
- orders
- order_items
- reviews
- messages
- notifications
- complaints

Unique Twist

Marketplace Trust Score

Each seller has a **Trust Score** (0–100) that is automatically updated based on marketplace activity.

- Every completed order increases the score by **+1**.
- Every 5-star review increases the score by **+2**.
- Every cancelled order decreases the score by **-2**.
- Every verified complaint decreases the score by **-5**.

The Trust Score is displayed on every seller's store page using a colored badge:

- **90–100**: Trusted Seller (Gold)
- **70–89**: Verified Seller (Green)
- **50–69**: Standard Seller (Blue)
- **Below 50**: New/Needs Improvement (Gray)

This feature encourages sellers to maintain good service quality.

Evaluation Rubric

Criterion	Marks
Functionality	40
User Interface	15
Database Design	15
Code Quality	10
Security	10
Presentation	10
Total	100

Submission Guidelines

- Submit complete source code.
- Submit SQL export file.
- Include a README describing setup instructions.
- Every group member must participate during demonstration.
- Projects should be original. Copying from online sources is prohibited.
- Use meaningful variable names and proper code organization.